Lesson notes · for you

Learn from reward

AI · University · Lesson 1

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- The reward is at the end
- Early steps must learn too

Activity: In the lab

1. Press “Watch the robot”: it has learned nothing, so it is lost. Press “Practise” (200 trips), watch again, and follow the arrows to the charger.
2. Choose one square. Its shading is its value (darker: closer to a reward); its arrow is the policy, the move it makes there. Say both aloud.
3. On paper, work the reading’s update: the move into the charger becomes $0 + 0.5 \times (10 - 0) = 5$; from the square before it, the target is $0.9 \times 5 - 0.1 = 4.4$, so that move becomes $0.5 \times 4.4 = 2.2$.

Discussion

- Can a reward be wrong?

Check yourself

- Value is not policy. Finish the sentence.

Try it this week
Write a one-line reward that would teach the wrong behaviour.

Worksheet

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Name _____ Class _____ Date _____

1 **Are a value and a policy the same picture?**

- ◯ Yes, once the robot has stopped exploring
- ◯ Yes: they are two names for one map
- ◯ No: value is how good a square is; policy is which way to step there

2 **A move into the charger is worth 6. The robot makes it again: the target is 10 and the learning rate is 0.5. What is the move worth after this update?**

_____

3 **Match the word.**

Reward	• • • •	How good a place is
Value	• • • •	What to do there
Policy	• • • •	The number the world returns

4 **Order one learning step from reward.**

- ☐ Update the value of where you were
- ☐ Take an action
- ☐ See the new place and the reward

✎ **Where does the credit for reaching the charger have to travel?**

_____

_____

_____

Check your answers

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1 **No: value is how good a square is; policy is which way to step there**

Value is a number for each square: how good it is to be there.

2 **8; 8.0**

The value goes half way from 6 to the target 10: 6 + 0.5 × (10 − 6) = 8.

3 **Reward — The number the world returns**
Value — How good a place is
Policy — What to do there

A reward that looks sensible can still train a bad policy.

4 **Top to bottom: 3, 1, 2**

The update is last. It needs the reward.